A Case Study Report

on

“Airline Customer Segmentation using Clustering and Wine Quality Prediction using Regression”

for the course

Fundamentals of Data Science (22CSC59)

Batch 2 – Case Study 2 and Case Study 5

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Section: CSE-3

Year: III Year / V Semester

Subject: Fundamentals of Data Science (22CSC59)

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[2026–2027]

CASE STUDY 2
AIRLINE CUSTOMER SEGMENTATION

1. Problem Statement

East-West Airlines wants to know more about its frequent-flyer customers. The three key questions are how often customers fly, how they earn and use reward miles, and how they use the airline credit card. At present the airline treats all customers the same way, so it cannot give the right offer to the right customer.

Problem: use clustering to divide the customers into meaningful segments, compare K-means, hierarchical clustering and DBSCAN, and explain what each segment means for the business.

Objectives:

- Clean the data, handle outliers and scale the features.
- Explore the data with statistics and graphs to find patterns.
- Apply K-means, hierarchical clustering and DBSCAN, and try different parameter settings for each.
- Visualise the clusters and evaluate them with the silhouette score and the Davies–Bouldin index.
- Describe each customer segment and suggest business actions.

2. Methodology

2.1 Dataset

The dataset *EastWestAirlines.xlsx* has **3,999 customers and 12 columns**. All columns are numbers:

- **Balance:** miles available for award travel.
- **Qual_miles:** miles that count towards Topflight (elite) status.
- **cc1_miles, cc2_miles, cc3_miles:** miles earned with three types of credit card, coded 1–5 (1 = under 5,000 miles, 5 = over 50,000 miles).
- **Bonus_miles and Bonus_trans:** miles and number of transactions from non-flight bonuses.
- **Flight_miles_12mo and Flight_trans_12:** flight miles and number of flights in the last 12 months.
- **Days_since_enroll:** how long the customer has been a member.
- **Award?:** 1 if the customer has redeemed an award, otherwise 0.

2.2 Steps followed

- **Data cleaning:** there were no missing values, so no filling was needed. One duplicate row was removed and the ID column was dropped. cc2_miles and cc3_miles were

dropped because about 99% of customers have the same value (1) in them, so they cannot separate customers.

- **Outlier handling:** outliers were found with box plots. Two rules were compared. The IQR rule would remove 1,089 customers (27%), while the Z-score rule ($|z| > 3$) removes only 321 customers (8%). The Z-score rule was used so that real active customers are not thrown away. After this step, 3,677 customers remained.
- **Feature scaling:** clustering uses Euclidean distance. Balance is in lakhs while cc1_miles is only 1–5, so all 9 features were scaled with StandardScaler (mean 0, standard deviation 1).
- **Exploratory data analysis:** descriptive statistics, histograms, box plots, scatter plots, a correlation heatmap, a pair plot and PCA.
- **K-means:** places k centres and assigns each customer to the nearest one. k was chosen with the elbow method (WCSS for k = 1 to 10), the silhouette score and the Davies–Bouldin index.
- **Hierarchical clustering (agglomerative):** starts with each customer as its own cluster and keeps merging the two closest clusters. Single, complete, average and ward linkage were compared.
- **DBSCAN:** groups customers that lie in dense areas and marks isolated customers as noise. eps (0.6 to 1.4) and min_samples (5, 10 and 18) were tested.
- **Evaluation:** the silhouette score (−1 to 1, higher is better) and the Davies–Bouldin index (lower is better). The Rand score was used to check how much two algorithms agree.

3. Statistical Findings

3.1 Descriptive statistics

Table 2.1: Descriptive statistics of the numeric features (after outlier removal)

Feature	Mean	Median	Std. dev.	Skewness	% zeros
Balance	60,536	39,372	61,875	1.89	0.03
Qual_miles	39.4	0	248.4	7.09	97.0
Bonus_miles	14,416	6,000	18,232	1.58	12.6
Bonus_trans	10.5	11	8.3	0.60	12.6
Flight_miles_12mo	256.7	0	637.2	3.55	72.1
Flight_trans_12	0.79	0	1.83	3.28	72.1
Days_since_enroll	4,061	4,015	2,051	0.14	0.0

For most mileage features the mean is much bigger than the median and the skewness is high. This means most customers have small values and a few customers have very large values (right-skewed). **97%** of customers have no qualifying miles and **72%** took no flight in the last 12 months. Days_since_enroll is almost symmetric (skewness 0.14).

3.2 Outliers

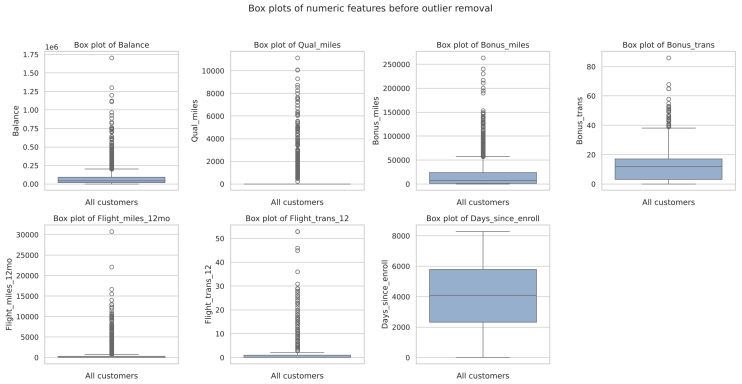


Figure 2.1: Box plots of numeric features before outlier removal

Interpretation: every mileage feature has many points above the upper whisker. A few customers have very large values, for example a Balance of up to 17 lakh miles. These outliers can pull the cluster centres towards them, so the most extreme 8% of customers were removed using the Z-score rule.

3.3 Univariate analysis

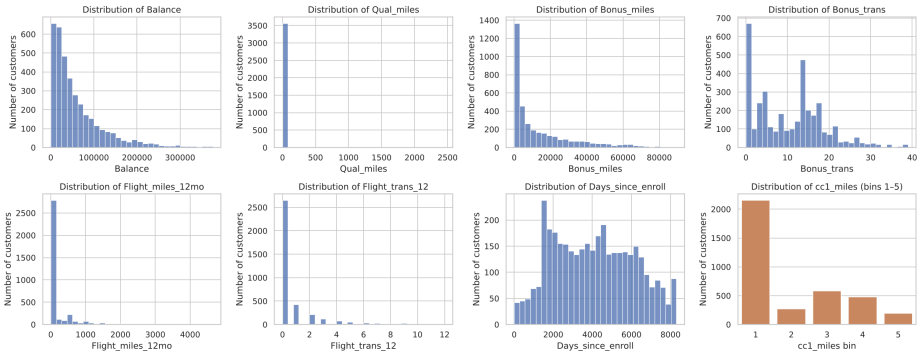


Figure 2.2: Distribution of each feature after outlier removal

Interpretation: most features are right-skewed, with a tall bar near zero and a long tail. Days_since_enroll is spread almost evenly, which shows that customers joined steadily over the years. Most customers are in card-miles bin 1 (under 5,000 miles).

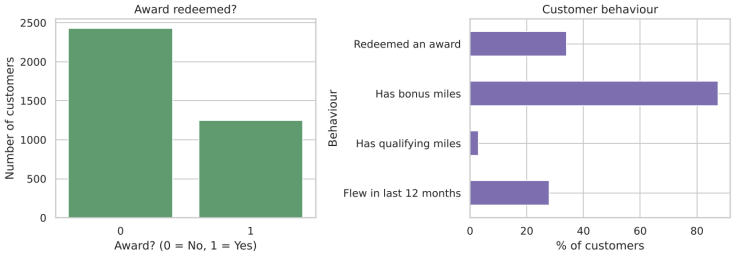


Figure 2.3: Award redemption and share of customers showing key behaviours

Interpretation: only **27.9%** of customers flew in the last 12 months and only **3.0%** have qualifying miles, but **87.4%** earned bonus miles. So most customers *earn* miles without *flying*. About **34%** have redeemed an award.

3.4 Bivariate and multivariate analysis

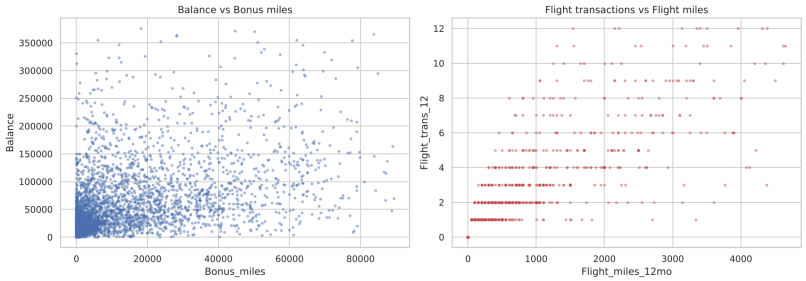


Figure 2.4: Scatter plots of Balance vs Bonus miles and Flight transactions vs Flight miles

Interpretation: balance increases as bonus miles increase, and the number of flights increases almost in a straight line with flight miles. Many points lie on the axes, i.e. customers with zero bonus miles or zero flights.

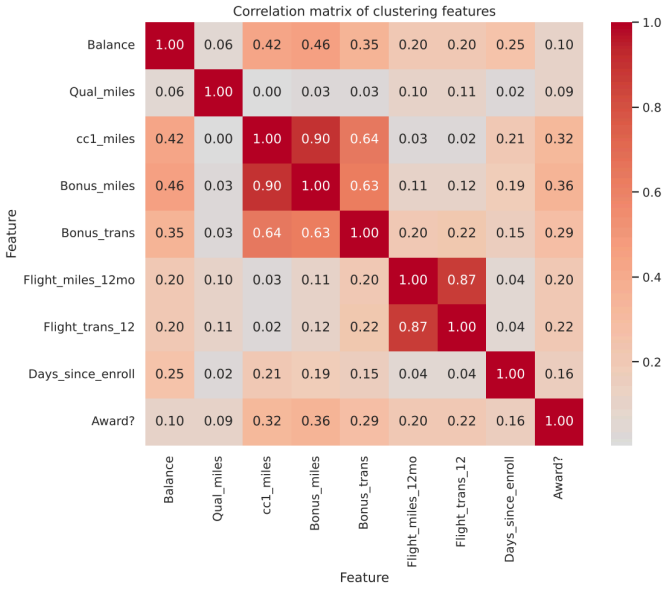


Figure 2.5: Correlation matrix of clustering features

Interpretation: credit-card miles and bonus miles are strongly related (r = 0.90), and flight miles and flight transactions are strongly related (r = 0.87). These two groups are only weakly related to each other (r ≈ 0.0–0.2). This means **earning miles** and **flying** are two separate behaviours. Award? is related more to bonus miles (r = 0.36) than to Balance (r = 0.10).

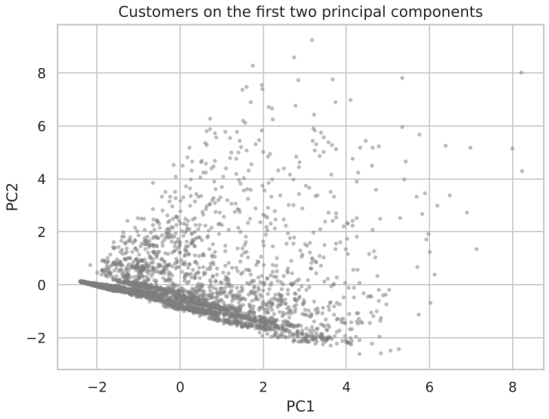


Figure 2.6: Customers plotted on the first two principal components

Interpretation: PCA was used only to draw the 9-feature data in 2-D. PC1 (35% of the variance) mainly measures card and bonus earning, and PC2 (20%) mainly measures flying. The plot shows a dense band of low-activity customers and a spread-out cloud of active customers.

Key findings: most customers earn miles but do not fly. The data is right-skewed, with a few heavy users. Earning and flying are the two main patterns along which customers can be grouped.

4. Model Evaluation

4.1 K-means clustering

Table 2.2: K-means results for different numbers of clusters (k)

k	WCSS	Silhouette score	Davies–Bouldin index
2	24,878	0.313	1.528
3	20,318	0.318	1.301
4	**17,608**	**0.331**	**1.163**
5	15,727	0.276	1.262
6	14,172	0.269	1.280
7	13,000	0.238	1.259
8	12,129	0.246	1.258



Figure 2.7: Elbow method, silhouette score and Davies–Bouldin index for different values of k

Interpretation: the WCSS curve drops quickly up to k = 4 and then flattens (the elbow). The silhouette score is highest at **k = 4 (0.331)** and the Davies–Bouldin index is lowest at **k = 4 (1.163)**. All three methods agree, so **k = 4** was chosen. The four clusters have 2,174, 1,190, 241 and 72 customers.

4.2 Hierarchical clustering

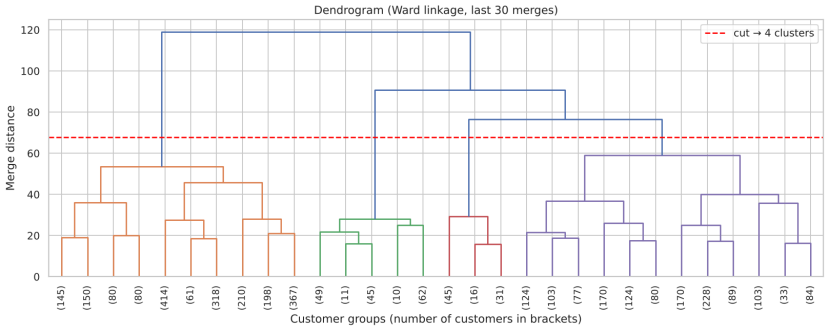


Figure 2.8: Dendrogram of hierarchical clustering (ward linkage)

Interpretation: the last three merges happen at a much larger distance than the others, so the data has a few big groups. Cutting the tree at the red line gives 4 clusters.

Table 2.3: Effect of the linkage criterion (k = 4)

Linkage	Silhouette score	Davies–Bouldin index	Cluster sizes
Single	0.464	0.356	3,673 / 2 / 1 / 1
Complete	0.479	1.097	3,478 / 105 / 85 / 9
Average	0.501	1.122	3,502 / 115 / 48 / 12
Ward	**0.288**	**1.251**	**2,023 / 1,385 / 177 / 92**

Interpretation: single, complete and average linkage show higher silhouette scores, but they put almost every customer into one giant cluster, leaving only a handful of customers in the others. This is called the *chaining effect*, and such a split is not useful. **Ward linkage** gives balanced clusters, so it was selected. A high silhouette score alone does not guarantee useful clusters; the cluster sizes must also be checked.

4.3 DBSCAN clustering

Table 2.4: DBSCAN results for different eps and min_samples

min_samples	eps	Clusters	Noise (%)	Silhouette score
5	0.6	21	36.3	0.043
5	1.0	3	15.8	0.268
10	0.6	10	45.0	0.174
10	0.8	4	29.9	0.274
10	**1.0**	**2**	**19.8**	**0.301**

min_samples	eps	Clusters	Noise (%)	Silhouette score
10	1.4	2	10.3	0.276
18	0.8	2	35.2	0.336
18	1.2	2	16.5	0.293

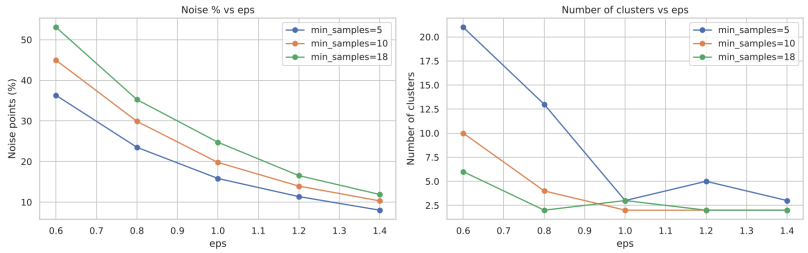


Figure 2.9: Effect of eps and min_samples on DBSCAN

Interpretation: a small eps marks many customers as noise and creates many tiny clusters (21 clusters and 36% noise at eps = 0.6). A large eps reduces noise but leaves only two clusters. A larger min_samples increases the noise. Among settings with at most 20% noise, **eps = 1.0 and min_samples = 10** gave the best silhouette score (0.301), with 2 clusters and 19.8% noise.

4.4 Visualisation of the clusters

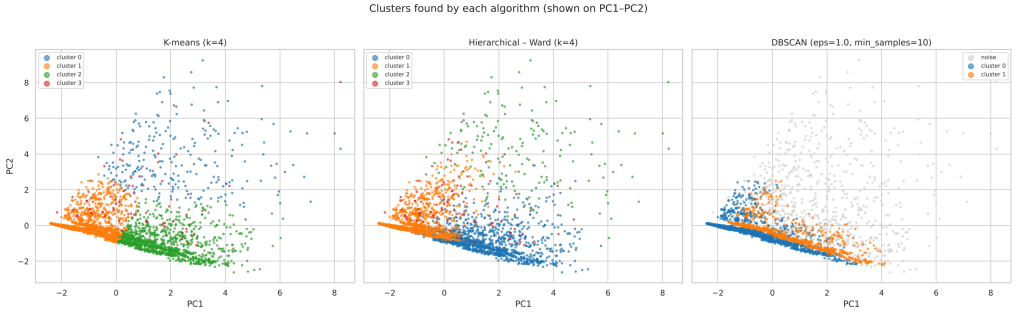


Figure 2.10: Clusters found by K-means, hierarchical clustering and DBSCAN (PC1 vs PC2)

Interpretation: K-means and ward give almost the same groups. The dense lower band is split into low-activity customers and card earners, and the spread-out upper area holds the frequent flyers and Topflight members. DBSCAN finds only two bands and marks most of the active customers as noise (grey).

4.5 Cluster analysis and interpretation

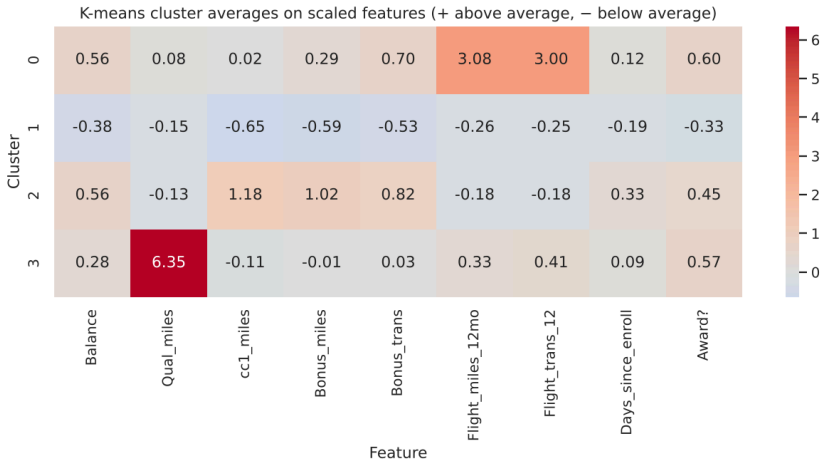


Figure 2.11: Average value of each feature in each K-means cluster (scaled)

Interpretation: red cells show where a cluster is above the average customer and blue cells show where it is below. Each cluster has a clear pattern: cluster 0 is high mainly on flight features, cluster 3 on qualifying miles, cluster 2 on card and bonus miles, and cluster 1 is below average on everything. This makes the clusters easy to name.

Table 2.5: Customer segments found by K-means (k = 4)

Segment	Customers	Avg. balance	Bonus miles	Flights (12 mo)	Redeemed
Low-activity members	2,174 (59%)	37,150	3,684	0.3	18%
Credit-card earners	1,190 (32%)	95,177	32,947	0.5	55%
Frequent flyers	241 (7%)	95,317	19,772	6.3	62%
Topflight qualifiers	72 (2%)	77,728	14,244	1.5	61%

- **Low-activity members (59%):** lowest balance, very few bonus miles, almost no flights and the newest members. Only 18% have redeemed an award.
- **Credit-card earners (32%):** high balance and the most bonus miles from the credit card (highest card bin), but they hardly fly. They collect miles without buying tickets.
- **Frequent flyers (7%):** about 6 flights in the last year and a high balance. 62% have redeemed an award.
- **Topflight qualifiers (2%):** the only group with qualifying miles (about 1,600 on average). They are the most valuable customers.

Hierarchical (ward) clusters: ward found almost the same four segments: card earners (1,385), low-activity members (2,023), frequent flyers (177) and Topflight qualifiers (92). 70 of the 72 Topflight qualifiers and 1,137 of the 1,190 card earners are in matching clusters. The **Rand score of 0.85** shows that the two methods agree for 85% of customer pairs, so the segments are stable.

DBSCAN clusters: DBSCAN split customers only by Award?. One cluster contains only customers who never redeemed an award and the other only customers who did. The 20% of customers marked as noise are the most valuable ones: all 72 Topflight qualifiers and 238 of the 241 frequent flyers. They are spread out (low density), so DBSCAN treats them as outliers.

4.6 Comparison of algorithms

Table 2.6: Comparison of the three clustering algorithms

Algorithm	Clusters	Noise	Silhouette ↑	Davies–Bouldin ↓
K-means (k = 4)	4	0%	**0.331**	**1.163**
Hierarchical – ward (k = 4)	4	0%	0.288	1.251
DBSCAN (eps = 1.0, min_samples = 10)	2	19.8%	0.301 (without noise)	1.534

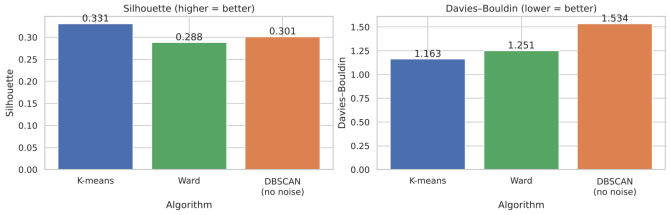


Figure 2.12: Comparison of silhouette score and Davies–Bouldin index

Interpretation: K-means has the **highest silhouette score (0.331)** and the **lowest Davies–Bouldin index (1.163)**, so it performs best. Ward is close behind. DBSCAN's silhouette score looks good only because its 20% noise points are left out, and its Davies–Bouldin index (1.534) is the worst. All scores are below 0.5, so the clusters are reasonably but not sharply separated, which is normal for real customer data.

5. Conclusion

K-means with k = 4 gave the best and most useful result. It divided the customers into **low-activity members, credit-card earners, frequent flyers and Topflight qualifiers**, and hierarchical clustering with ward linkage found almost the same groups. Single, complete and average linkage put nearly all customers into one cluster. DBSCAN treated the most valuable customers as noise, so it was not suitable for segmentation, although it is useful for spotting unusual customers.

Business insights: keep **Topflight qualifiers and frequent flyers** loyal with status benefits, encourage **credit-card earners** (who collect many miles but rarely fly) to book flights with special fare offers, and send low-cost re-activation offers to **low-activity members**.

Limitations: outlier removal also removed some Topflight customers, card miles are only bins (1–5), and results depend on the chosen k, eps and min_samples.

CASE STUDY 5
WINE QUALITY ANALYSIS AND PREDICTION

1. Problem Statement

The company currently checks wine quality using tasters (sensory assessment). This is slow and depends on personal opinion. Management wants to know whether laboratory (physicochemical) measurements can explain differences in wine quality and support a data-driven quality check.

Problem: find which laboratory measurements are most strongly related to wine quality, and build and evaluate a regression model that predicts the quality score from these measurements.

2. Methodology

2.1 Dataset

The red wine dataset (*winequality-red.csv*, from the UCI Machine Learning Repository) has **1,599 wines**. Each wine has **11 laboratory measurements** and a **quality** score from tasters on a 0–10 scale. The 11 measurements are fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. The zip file also contains white wines, but only red wine was used because red and white wines have different chemistry.

2.2 Steps followed

- **Data check:** there were no missing values. The 240 duplicate rows were kept because they are likely the same wine tested more than once.
- **Statistics (Task 1):** mean, median, mode, standard deviation, skewness, coefficient of variation (CV) and Pearson correlation.
- **EDA (Task 2):** a histogram of quality, a box plot of alcohol by quality and a correlation heatmap.
- **Model (Task 3):** multiple linear regression, quality = $\beta_0 + \beta_1 x_1 + \ldots + \beta_{11} x_{11}$, using all 11 measurements. The data was split into 80% training (1,279 wines) and 20% testing (320 wines) with random_state = 42. The model was evaluated with MAE, MSE, RMSE and R^2.
- **Multicollinearity:** checked with the correlation matrix and the variance inflation factor (VIF). Features were standardised to compare the coefficients.

3. Statistical Findings

3.1 Descriptive statistics (Task 1a)

Table 5.1: Statistics of alcohol, volatile acidity and residual sugar

Statistic	Alcohol	Volatile acidity	Residual sugar
Mean	10.42	0.528	2.54
Median	10.20	0.520	2.20
Mode	9.50	0.600	2.00
Standard deviation	1.07	0.179	1.41
Skewness	0.86	0.67	4.54

- **Alcohol:** mean (10.42) > median (10.20) > mode (9.5), so it is moderately right-skewed. Most wines have about 9.5–10.5% alcohol, and a few stronger wines form a tail.
- **Volatile acidity:** the mean and median are almost equal, so it is only slightly right-skewed (0.67).
- **Residual sugar:** highly right-skewed (4.54). Most red wines are dry (about 2 g/dm³) and only a few are very sweet.

3.2 Relative variability (Task 1b)

The three attributes have different units, so their standard deviations cannot be compared directly. The **coefficient of variation** (CV = standard deviation ÷ mean × 100) shows the spread as a percentage of the mean.

Table 5.2: Coefficient of variation

Attribute	Standard deviation	Mean	CV
Residual sugar	1.41	2.54	**55.5%**
Volatile acidity	0.179	0.528	33.9%
Alcohol	1.07	10.42	10.2%

Residual sugar has the greatest relative variability (CV = 55.5%) because of the few very sweet wines. Alcohol is the most stable (10.2%).

3.3 Alcohol and quality (Task 1c)

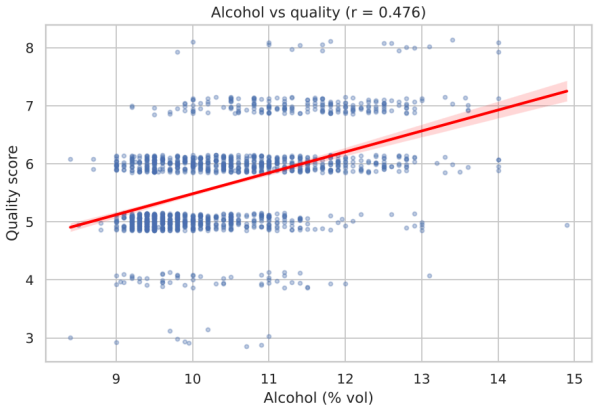


Figure 5.1: Alcohol content vs quality score

Interpretation: Pearson's r = **+0.476**. The **direction is positive** (wines with more alcohol usually get higher scores) and the **strength is moderate** (between 0.3 and 0.5). The p-value is almost zero, so the relationship is statistically significant. However, $r^2 = 0.23$, so alcohol alone explains only about 23% of the differences in quality.

3.4 Distribution of quality (Task 2a)

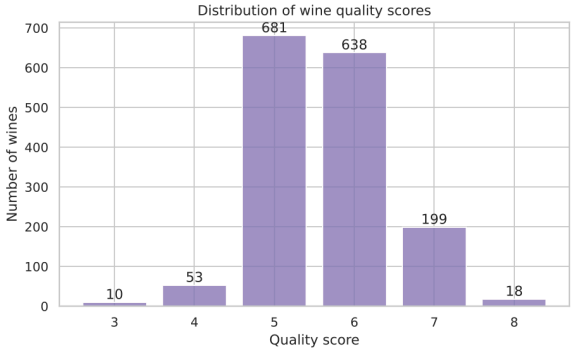


Figure 5.2: Distribution of wine quality scores

Interpretation: only scores 3 to 8 appear. Scores 5 (681 wines) and 6 (638 wines) together make up about **82%** of all wines. Only 63 wines score 3–4 and only 18 score 8. The distribution has one peak, but the data is **imbalanced**, so a model will find it hard to predict very poor or very good wines.

3.5 Alcohol by quality (Task 2b)

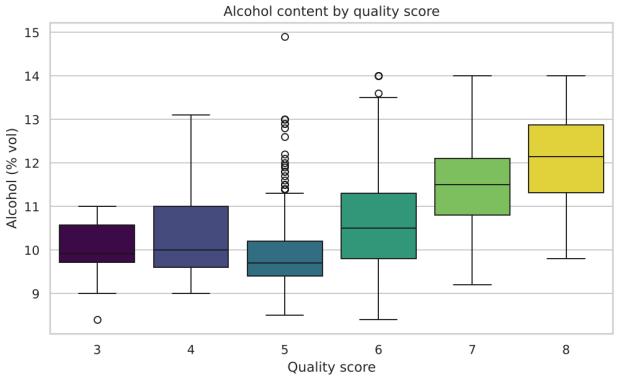


Figure 5.3: Alcohol content by quality score

Interpretation: median alcohol stays at about 10% for quality 3 to 5, then rises clearly for better wines. Quality 5 has the most outliers (30 wines): these have high alcohol (up to 14.9%) but still scored only 5, which shows that alcohol alone does not decide quality.

Table 5.3: Alcohol content for each quality score

Quality	Wines	Median alcohol (%)	Mean alcohol (%)	Outliers
3	10	9.93	9.96	1
4	53	10.00	10.27	0
5	681	9.70	9.90	30
6	638	10.50	10.63	5
7	199	11.50	11.47	0
8	18	12.15	12.09	0

3.6 Correlation heatmap (Task 2c)

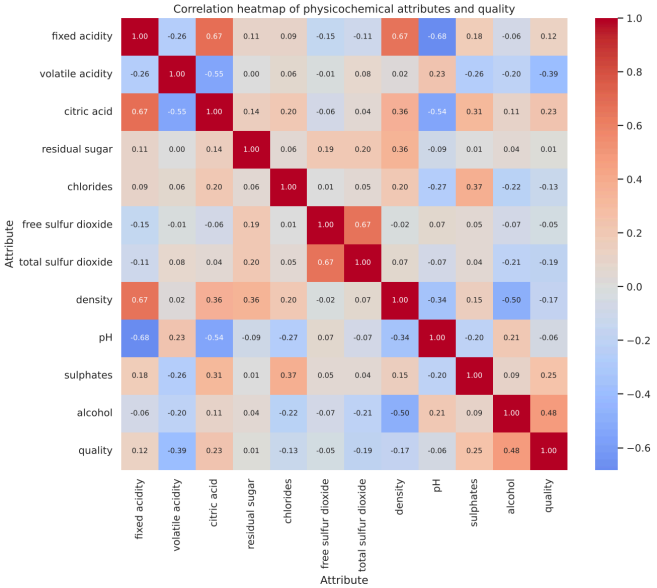


Figure 5.4: Correlation heatmap of the measurements and quality

Interpretation: the last row shows how each measurement relates to quality. Some measurements are also strongly related to each other: fixed acidity with citric acid (+0.67), with density (+0.67) and with pH (−0.68), and free with total sulfur dioxide (+0.67). Such relationships can cause **multicollinearity** in the regression model.

3.7 Attributes most associated with quality (Task 2d)

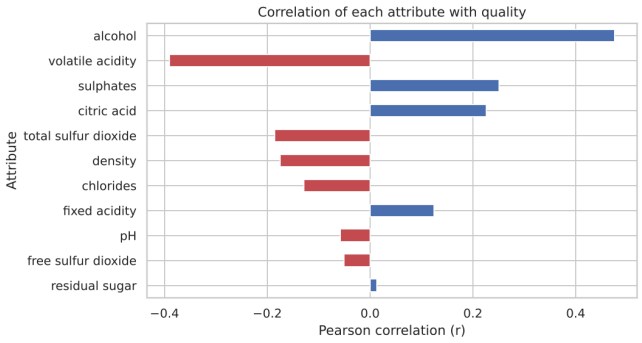


Figure 5.5: Correlation of each measurement with quality

Interpretation: the two measurements with the largest absolute correlation are **alcohol (r = +0.476, positive)** and **volatile acidity (r = −0.391, negative)**. More alcohol goes with higher quality. Volatile acidity is mostly acetic acid, which gives a vinegar taste, so more of it goes with lower quality. Residual sugar has almost no relationship with quality (r = 0.01).

4. Model Evaluation

4.1 Model training (Tasks 3a and 3b)

A multiple linear regression model was trained on 1,279 wines (80%) and tested on 320 unseen wines (20%) with random_state = 42. The intercept is 14.36. Some of the learned coefficients are alcohol +0.28 per 1% alcohol, volatile acidity −1.00 per g/dm³ and sulphates +0.84 per g/dm³. These raw coefficients cannot be compared with each other because the measurements use different units.

4.2 Model performance (Task 3c)

Table 5.4: Performance of the model on the test set (320 wines)

Metric	Value	Meaning
MAE	0.504	On average, the prediction is half a point away from the real score
MSE	0.390	Average of the squared errors (big errors count more)
RMSE	0.625	Typical error is about 0.6 quality points
R^2	0.403	The model explains about 40% of the variation in quality

An **R^2 of 0.40** means the laboratory measurements explain part of the quality, but 60% of the differences still depend on other things, such as grape type, age and the taster's opinion. An average error of half a point is large when most wines score 5 or 6.

4.3 Most influential predictors and multicollinearity (Task 3d)

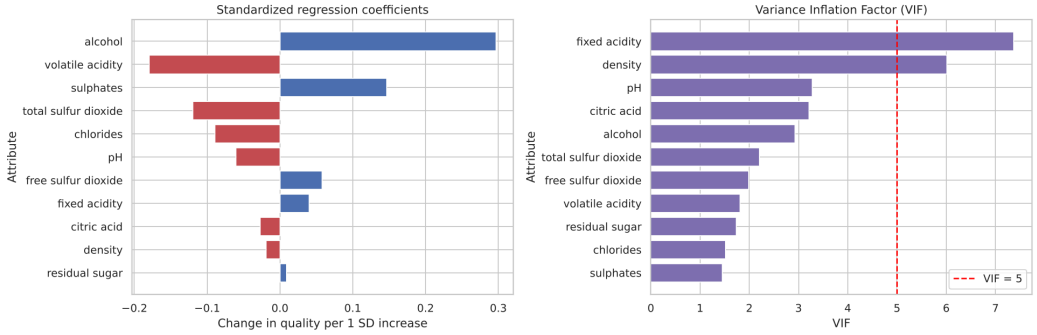


Figure 5.6: Standardized regression coefficients and VIF of each measurement

Interpretation: after scaling all measurements to the same scale, the two most influential predictors are **alcohol (+0.30)** and **volatile acidity (−0.18)**, followed by sulphates (+0.15) and total sulfur dioxide (−0.12).

Table 5.5: Standardized coefficients and VIF (top predictors and high-VIF features)

Measurement	Standardized coefficient	VIF	Multicollinearity
Alcohol	+0.297	2.93	Low
Volatile acidity	−0.179	1.81	Low
Sulphates	+0.147	1.46	Low
Total sulfur dioxide	−0.120	2.20	Low
Fixed acidity	+0.040	7.38	Moderate (VIF > 5)
Density	−0.019	6.02	Moderate (VIF > 5)

Multicollinearity was checked before interpreting the effects. **Fixed acidity (VIF 7.4) and density (VIF 6.0)** are strongly related to citric acid, pH and alcohol, so their coefficients are not reliable. **Alcohol and volatile acidity** have low VIF, so their effects can be trusted:

- If alcohol increases by one standard deviation (about 1.07%), the predicted quality rises by about **0.30 points**.
- If volatile acidity increases by one standard deviation (about 0.18 g/dm³), the predicted quality falls by about **0.18 points**.

4.4 Predicted vs actual quality (Task 3e)

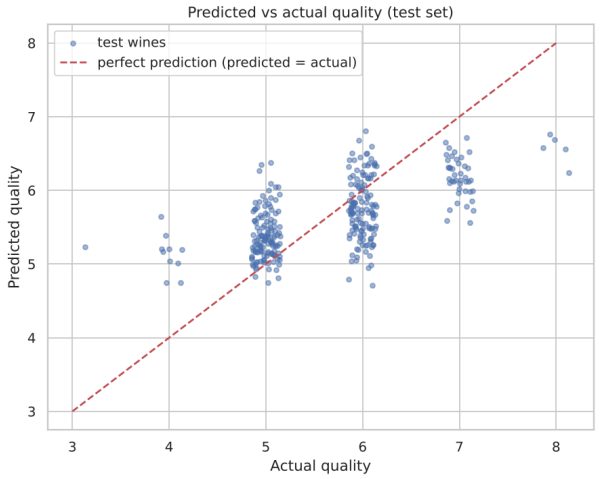


Figure 5.7: Predicted vs actual quality on the test set

Interpretation: if the model were perfect, all points would lie on the red line. Instead, predictions stay between 4.7 and 6.8 while the actual scores go from 3 to 8.

Table 5.6: Average predicted score for each actual score (test set)

Actual quality	**3**	**4**	**5**	**6**	**7**	**8**
Test wines	1	10	130	132	42	5
Average prediction	5.23	5.14	5.38	5.76	6.18	6.56

Limitations of the model:

- It **over-predicts poor wines** (an actual 3 or 4 is predicted at about 5.2) and **under-predicts excellent wines** (an actual 8 is predicted at about 6.6), so it pulls every wine towards the average.
- Quality is a whole number, but linear regression predicts decimal values.
- There are very few poor and excellent wines to learn from.
- The relationship is not perfectly straight; for example, alcohol hardly changes between quality 3 and 5.

5. Conclusion

Laboratory measurements can explain part of wine quality. **Alcohol** (positive) and **volatile acidity** (negative) are the most important measurements, both in the correlation analysis and in the regression model. The model reached **R^2 = 0.40, RMSE = 0.62 and MAE = 0.50** on the test set.

5.1 Recommended screening measurements (Task 4a)

- **Alcohol:** it has the strongest positive link with quality, so it helps to identify good wines.
- **Volatile acidity:** it has the strongest negative link with quality, so it helps to catch poor or faulty wines (vinegar taste).

5.2 Can the model replace sensory evaluation? (Task 4b)

No. The model explains only 40% of the variation in quality ($R^2 = 0.40$). Its average error of half a point (MAE = 0.50, RMSE = 0.62) is the same as the gap between an average wine (5) and a good wine (6), and 82% of wines are exactly 5 or 6. It also rates poor wines as average, so faulty wines could pass. The model should be used only as a **first screening step**, for example to send wines with high volatile acidity or a low predicted score for tasting first, while tasters make the final decision.

5.3 Limitations (Task 4c)

- **Dataset:** quality is a subjective score given by tasters. The data covers only one type of red wine, has very few poor or excellent wines, and has no information on grape type, age or price.
- **Approach:** linear regression assumes straight-line relationships and treats a whole-number score as continuous. Some measurements are correlated with each other, and the results come from a single 80/20 split.